

phase 1

28/02/2026

## → Number system Conversion

1. Convert the following decimal numbers to binary:

a) 18

b) 45

$$\begin{array}{r} 2 \overline{) 18} \phantom{0} \\ \underline{2 \phantom{0}} \phantom{0} \\ 18 \phantom{0} \\ \underline{16} \phantom{0} \\ 2 \phantom{0} \\ \underline{2} \phantom{0} \\ 0 \phantom{0} \end{array}$$

$$a = 10010$$

$$\begin{array}{r} 2 \overline{) 45} \phantom{1} \\ \underline{2 \phantom{0}} \phantom{1} \\ 22 \phantom{1} \\ \underline{2 \phantom{0}} \phantom{1} \\ 11 \phantom{1} \\ \underline{10} \phantom{1} \\ 1 \phantom{1} \\ \underline{1} \phantom{1} \\ 0 \phantom{1} \end{array}$$

$$b = 101101$$

2) Convert the following binary numbers to decimal.

a) 10101

$$\begin{array}{r} 10101 \\ 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ \hline 16 \quad 0 \quad 4 \quad 0 \quad 1 \\ \hline \end{array}$$

$$= (21)_{10}$$

b) 110011

$$\begin{array}{r} 110011 \\ 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ \hline 32 \quad 16 \quad 0 \quad 0 \quad 2 \quad 1 \\ \hline \end{array}$$

$$b = (51)_{10}$$



3) Convert the following decimal numbers to octal.

a) 74

$$\begin{array}{r} 8 \overline{) 74} \phantom{00} 2 \\ 8 \overline{) 91} \\ 1 \end{array}$$

a = 112

b) 255

$$\begin{array}{r} 8 \overline{) 255} \phantom{00} 7 \\ 8 \overline{) 317} \\ 3 \end{array}$$

b = 377

4) Convert the following octal numbers to decimal.

a)  $(127)_8$

$$\begin{array}{r} 1 \quad 2 \quad 7 \\ 8^2 \quad 8^1 \quad 8^0 \\ \hline 64 + 16 + 7 \end{array}$$

a = 87

b)  $(345)_8$

$$\begin{array}{r} 3 \quad 4 \quad 5 \\ 8^2 \quad 8^1 \quad 8^0 \\ \hline 192 + 32 + 5 \end{array}$$

b = 229

5) Convert the following Hexadecimal numbers to decimal.

a)  $(1A)_{16}$

$$\begin{array}{r} 1 \quad A \\ 16^1 \quad 16^0 \\ \hline 16 + 10 \end{array}$$

a = 26

b)  $(2F)_{16}$

$$\begin{array}{r} 2 \quad F \\ 16^1 \quad 16^0 \\ \hline 32 + 15 \end{array}$$

b = 47



6) Convert  $(255)_{10}$  to binary, octal & hexa-decimal

→ binary:-

$$\begin{array}{r}
 2 \overline{) 255} \\
 \underline{2} \phantom{00} 127 \\
 2 \overline{) 127} \\
 \underline{2} \phantom{00} 63 \\
 2 \overline{) 63} \\
 \underline{2} \phantom{00} 31 \\
 2 \overline{) 31} \\
 \underline{2} \phantom{00} 15 \\
 2 \overline{) 15} \\
 \underline{2} \phantom{00} 7 \\
 2 \overline{) 7} \\
 \underline{2} \phantom{00} 3 \\
 2 \overline{) 3} \\
 \underline{2} \phantom{00} 1
 \end{array}$$

$$\underline{(11111111)_2}$$

→ octal

$$\begin{array}{r}
 8 \overline{) 255} \\
 \underline{8} \phantom{00} 31 \\
 3
 \end{array}$$

$$\underline{(377)_8}$$

→ Hexa-decimal

$$\begin{array}{r}
 16 \overline{) 255} \\
 \underline{16} \phantom{00} 15 \\
 15
 \end{array}$$

$$\underline{(FF)_{16}}$$

7) Convert  $(1101011)_2$  to octal & Hexa-decimal

→ binary to octal

$$(1101011)_2 \rightarrow (?)_8$$

$$\begin{array}{r}
 1 \phantom{0} 1 \phantom{0} 0 \phantom{0} 1 \phantom{0} 0 \phantom{0} 1 \phantom{0} 1 \\
 2^6 \phantom{0} 2^5 \phantom{0} 2^4 \phantom{0} 2^3 \phantom{0} 2^2 \phantom{0} 2^1 \phantom{0} 2^0
 \end{array}$$

$$\underline{64 \phantom{00} 32 \phantom{00} 0 \phantom{00} 8 \phantom{00} 0 \phantom{00} 2 \phantom{00} 1}$$

$$= 107$$

$$\begin{array}{r}
 8 \overline{) 107} \\
 \underline{8} \phantom{00} 13 \\
 5
 \end{array}$$

$$\underline{= 153}$$

$$(1101011)_2 \rightarrow (153)_8$$



→ Binary to Hexadecimal

$$(1101011) \rightarrow (6B)_{16}$$

$$\begin{array}{r} 1101011 \\ 2^6 \ 2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0 \\ \hline 107 \end{array}$$

$$\begin{array}{r} 16 \overline{) 107} \ 11 \\ \underline{6} \end{array}$$

$$\underline{26B}$$

8) Convert  $(3A5)_{16}$  to binary & decimal.

→ Hexadecimal to binary

$$(3A5)_{16} \rightarrow (\quad)_2$$

$$\begin{array}{r} 3 \ A \ 5 \\ 16^2 \ 16^1 \ 16^0 \\ \hline 268 + 160 + 5 \\ = 933 \end{array}$$

$$\begin{array}{r} 2 \overline{) 933} \ 1 \\ 2 \overline{) 466} \ 0 \\ 2 \overline{) 233} \ 1 \\ 2 \overline{) 116} \ 0 \\ 2 \overline{) 58} \ 0 \\ 2 \overline{) 29} \ 1 \\ 2 \overline{) 14} \ 0 \\ 2 \overline{) 7} \ 1 \\ 2 \overline{) 3} \ 1 \\ 1 \end{array}$$

$$\underline{\underline{(1110100101)_2}}$$



→ Hexadecimal to decimal

$$(3A5)_{16} \rightarrow \underline{(933)_{10}}$$

$$\begin{array}{r} 3 \quad A \quad 5 \\ 16^2 \quad 16^1 \quad 16^0 \\ \hline 933 \\ \hline \end{array}$$